

House Price Prediction Project

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1 Project Overview

This project builds a machine learning pipeline to predict **house prices** using the Ames Housing dataset. The dataset provides detailed attributes of residential homes (such as lot size, neighborhood, quality ratings, number of rooms). The goal is to predict the target variable **SalePrice** for unseen houses.

2 How It Works

The project follows these steps:

1. Data Preprocessing:

- Dropped irrelevant and very sparse features (**Alley**, **PoolQC**, **MiscFeature**).
- Handled missing values by filling numeric columns with median and categorical columns with mode.
- Applied ordinal mappings for quality-related attributes such as **ExterQual**, **KitchenQual**, **GarageQual**.
- Performed one-hot encoding for nominal categorical features like **Neighborhood**, **MSZoning**, **SaleType**.
- Created new engineered features:
 - **total_area** = sum of first floor, second floor, and basement
 - **house_age** = year sold – year built + 1
 - **garage_size_per_car**
 - **total_bathroom** = full + (1/2)*half bathrooms
 - **house** = composite score combining area, condition, and kitchen

2. Exploratory Data Analysis (EDA):

- Distribution of **SalePrice**
- Correlation of features with target
- Scatter plots to observe feature-target relationships

3. Model Training:

Several regression models were trained and optimized using **GridSearchCV** or **RandomizedSearchCV**:

- ElasticNet Regression (with scaling options)
 - Random Forest Regressor
 - XGBoost Regressor (GPU-accelerated)
 - Support Vector Regression (SVR)
4. **Ensembling:** Instead of a simple model, final predictions are generated by **weighted blending**. The inverse of RMSE (on validation sets) was used as model weights. Thus, better-performing models contribute more to the final output.
 5. **Submission:** The blended predictions are saved to `submission.csv`, ready for evaluation.

3 Technologies Used

- **Python Libraries:**
 - `pandas`, `numpy` for data processing
 - `scikit-learn` for preprocessing, pipelines, model training
 - `xgboost` for gradient boosting
 - `seaborn`, `matplotlib` for visualization
- **Dataset:**
 - `train.csv` (with target `SalePrice`)
 - `test.csv` (test data for predictions)
 - `data_description.txt` (feature definitions)

4 Future Improvements

Potential directions to extend this work:

- Apply more advanced feature engineering (interaction terms, polynomial features).
- Use stronger ensemble methods like `StackingRegressor` with meta-learners instead of weighted blending.
- Perform more rigorous hyperparameter optimization using Bayesian Optimization.
- Explore deep learning methods (e.g., feedforward neural networks).
- Deploy the model as a web service for real-time house price prediction.